

What Variables Do We Want to Include

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- Often have a large number of variables included in an observational dataset
- We have discussed various methods to estimate the causal effect which rely on fitting regression models without a lot of discussion on which variables to include or not include
- The only thing that we have definitively stated is that X needs to ensure that that $\{Y^1, Y^0\}$ are independent of A conditioned on X

- In some applications it makes sense to draw the causal structure/relationship among variables (regardless of whether they are measured)
- Arrows point in the direction of causal effect
- Known as a directed acyclic graph (“Directed:” edges imply a direction. “Acyclic:” no cycles, a variable cannot cause itself, either directly or through another variable.)

Example DAG

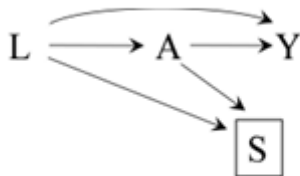


Figure 6.9

Figure 1: caption

- There is a lot of theory relating probability theory (conditional independence) to DAGs with some unusual terms (d-separation, collider, etc.)
- In particular, IF you have drawn the correct DAG, DAG will tell you which variables you would need to adjust for (causal reasoning) and which you should not
- In my experience, can spend a lot of time learning the “rules” of DAGs without a lot of benefit

- Ongoing work to be able to learn the DAG from just data alone
- May not represent causal relationships without additional assumptions

Some Simple Rules

To understand causal effect of A on Y

- Never adjust for variables which come after treatment decision or when randomization would have occurred. These are (intermediate) outcomes
- Do not care about the relationship between pre-treatment variables
- This leaves four types of variables

- 1) Common causes of A and Y
- 2) Causes of Y alone
- 3) Causes of A alone
- 4) Causes of neither A or Y

Causal Inference: Target Trial

- Throughout the course we have thought about trying to use analytical tools to mimic the randomized trial that we would have liked to perform (target trial)
- Framework works well for interventions which you can control (pharmaceutical treatment, surgery, behavioral intervention, income, etc.) and at what time point they are delivered
- Part of specifying target trial is the inclusion/exclusion criteria, intervention (type, intensity/dosage, timing), follow-up and outcomes
- Framework leaves out many factors which we may be interested in (particularly race/ethnicity, gender, etc.)

Causal Inference: Non-Target Trial

- Causal methods are useful to create balance among certain characteristics between different groups
- Important to think of what characteristics AT WHAT TIME POINT we want balanced
- In other words, WHEN would you have done the randomization - Balancing on different characteristics at different time points gives insight on when differences emerge
- For example, ER doctor looking at outcomes by race/ethnicity for those presenting with chest pain can or cannot balance on laboratory values at presentation

The Studies

The paper reported on two related studies. The first was a pre-registered meta-analysis of field studies looking at gender bias in hiring decisions. Field studies are studies where fake job applications are submitted to real companies, with everything about the applications held constant except the variable of interest - in this case, the sex of the applicant. By counting the number of callbacks received for female vs. male applications, researchers can measure ambient levels of gender bias. Schaerer and colleagues set out to meta-analyze all the field studies they could get their hands on - that is, to combine them all into one giant study. They ended up with 85 studies spanning 26 countries, 44 years, and nearly 400,000 fake applications. The studies covered male-dominated professions (e.g., engineer, truck driver), female-dominated professions (e.g., HR professional, nurse), and gender-balanced professions (e.g., graphic designer, lab worker).

Figure 2: caption

Some Simple Rules

- Which of the 4 types of variables do we want to include in regression adjustment?
- Simulation!

```
results <- NULL
num.sim <- 25
sample.size <- 1000
set.seed(1101985)
for(i in 1:num.sim){
  X1 <- rnorm(sample.size, mean = 0, sd = 1)
  X2 <- rnorm(sample.size, mean = 0, sd = 1)
  X3 <- rnorm(sample.size, mean = 0, sd = 1)
  X4 <- rnorm(sample.size, mean = 0, sd = 1)

  #generate data
  Y1 <- 1 + X1 + X2 + rnorm(sample.size, mean = 0, sd = 1)
  Y0 <- X1 + X2 + rnorm(sample.size, mean = 0, sd = 1)
```